

Adaptation Experiment 2: Retrieval Baselines

1 Signals and leakage-free retrieval

For query date s , the exact window convention is

$$X_s = (s - L, s], \quad Y_s = (s, s + H].$$

Let $V = f(X_s)$ be the vanilla backbone forecast and let C be the same backbone conditioned on retrieved examples. For neighbor j , let r_j be its last look-back date, Y_j its observed future, and $N_j = f(X_{r_j})$ its vanilla forecast. All neighbor values are transferred to the query's level and scale before use. Distance weights are

$$w_j = \frac{\exp(-\tilde{d}_j)}{\sum_{\ell=1}^k \exp(-\tilde{d}_\ell)}, \quad \hat{Y}_h = \sum_{j=1}^k w_j Y_{j,h}.$$

In period-aligned retrieval, the last dates of the look-backs satisfy $s - r_j \equiv 0 \pmod{P}$. In fixed mode every (X_{r_j}, Y_{r_j}) lies wholly inside T_0 . In online mode a neighbor is eligible only when $r_j + H \leq s$, so its complete future is observed by query time. A retrieved future may overlap the query look-back, which is already observed, but it can never overlap Y_s . These conditions prevent target leakage.

Experimental goal. Compare all useful direct combinations of the available information V, C, Y_j, N_j before introducing a neural adapter.

2 Anchored models

Every fitted forecast is written directly as V + linear signals. There is no intercept and no design based only on differences such as $Y_j - N_j$. The word “residual” below is only the agreed name for the model containing both Y_j and N_j .

A *shared* model has one coefficient vector used at every horizon. A *horizon* model fits H independent regressions. Each of those H regressions has the same number of coefficients as the shared regression; the total coefficient count is therefore multiplied by H .

Artifact name	Forecast at horizon h	Coefficients
vanilla	V_h	0
context_forecast	C_h	0
aggr-y	$\hat{Y}_h$	0
y_mean	$k^{-1} \sum_j Y_{j,h}$	0
aggr_y_mix_shared	$(1 - \lambda)V_h + \lambda\hat{Y}_h$	1
aggr_y_mix_horizon	$(1 - \lambda_h)V_h + \lambda_h\hat{Y}_h$	1 per h
context_ridge_shared	$V_h + aV_h + bC_h$	2
context_ridge_horizon	$V_h + a_hV_h + b_hC_h$	2 per h
aggr_y_ridge_shared	$V_h + aV_h + b\hat{Y}_h$	2
aggr_y_ridge_horizon	$V_h + a_hV_h + b_h\hat{Y}_h$	2 per h
y_ridge_shared	$V_h + aV_h + \sum_j b_j Y_{j,h}$	$1 + k$
y_ridge_horizon	$V_h + a_hV_h + \sum_j b_{j,h} Y_{j,h}$	$1 + k$ per h
cov_y_ridge_shared	$V_h + aV_h + bC_h + \sum_j c_j Y_{j,h}$	$2 + k$
cov_horizon_ridge_shared	$V_h + aV_h + bC_h + c\hat{Y}_h$	3
cov_horizon_ridge_horizon	$V_h + a_hV_h + b_hC_h + c_h\hat{Y}_h$	3 per h
residual_ridge_shared	$V_h + aV_h + \sum_j b_j Y_{j,h} + \sum_j c_j N_{j,h}$	$1 + 2k$
residual_ridge_horizon	$V_h + a_hV_h + \sum_j b_{j,h} Y_{j,h} + \sum_j c_{j,h} N_{j,h}$	$1 + 2k$ per h

Artifact name	Forecast at horizon h	Coefficients
full_ridge_shared	$V_h + aV_h + bC_h + \sum_j c_j Y_{j,h} + \sum_j d_j N_{j,h}$	$2 + 2k$
full_ridge_horizon	$V_h + a_h V_h + b_h C_h + \sum_j c_{j,h} Y_{j,h} + \sum_j d_{j,h} N_{j,h}$	$2 + 2k$ per h

The two lambda models are the only retained convex mixtures. Their least-squares estimate is projected onto $[0, 1]$ with $\lambda \leftarrow \min(1, \max(0, \lambda_{\text{LS}}))$; the horizon variant does this independently for each λ_h . Ridge coefficients are unconstrained.

3 Model-local split and validation protocol

Extraction fixes only

$$T_0 \text{ (30\%),} \quad T_1 + T_2 \text{ (50\%),} \quad T_3 \text{ (20\%),}$$

and writes **adapt** and **eval** payloads. For these baselines, **adapt** is split chronologically by whole query dates, with its final 20% assigned to T_2 by default.

1. Accumulate exact float64 sufficient statistics on T_1 .
2. For every ridge family, fit every $\alpha \in \{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ and select the value with lowest T_2 nMSE.
3. Refit that family on all of $T_1 + T_2$ using the selected α .
4. Fit the lambda models directly on $T_1 + T_2$ because they have no regularization hyperparameter to validate.
5. Score every selected model once on the complete untouched T_3 .

Features are RMS-standardized without centering, so a zero coefficient still means no added signal. Chunked sufficient statistics change memory usage, not the fitted objective. Optional fit-only sample caps never subsample final T_3 scoring. Optimistic **eval_fit** variants remain available behind an explicit flag for appendix diagnostics and are disabled by default.

Code: `src/experiments/extraction.py`; `src/adaptors/baselines/evaluate.py`.

Launchers: `extraction.slurm`; `baselines.slurm`.

4 Interpretation

aggr_y tests analog forecasting directly; **context** tests whether the backbone can consume retrieved context. The individual- Y models test whether learned neighbor-specific weights beat the fixed distance aggregation. **residual** tests whether neighbor forecasts add information about neighbor reliability, and **full** is the complete V, C, Y_j, N_j model. All claims must be based on the same complete T_3 queries and compared with V .